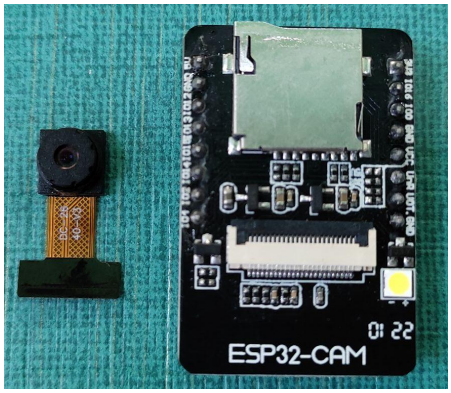
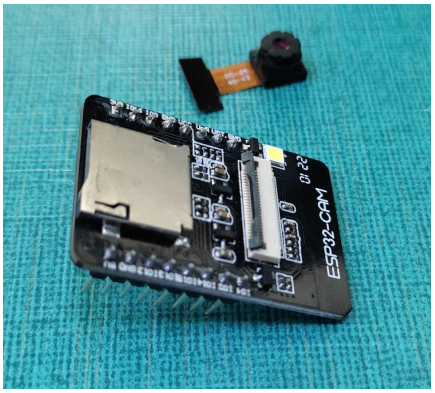
HARDWARE SETUP

To install the hardware and make it ready for firmware upload, following steps are required:

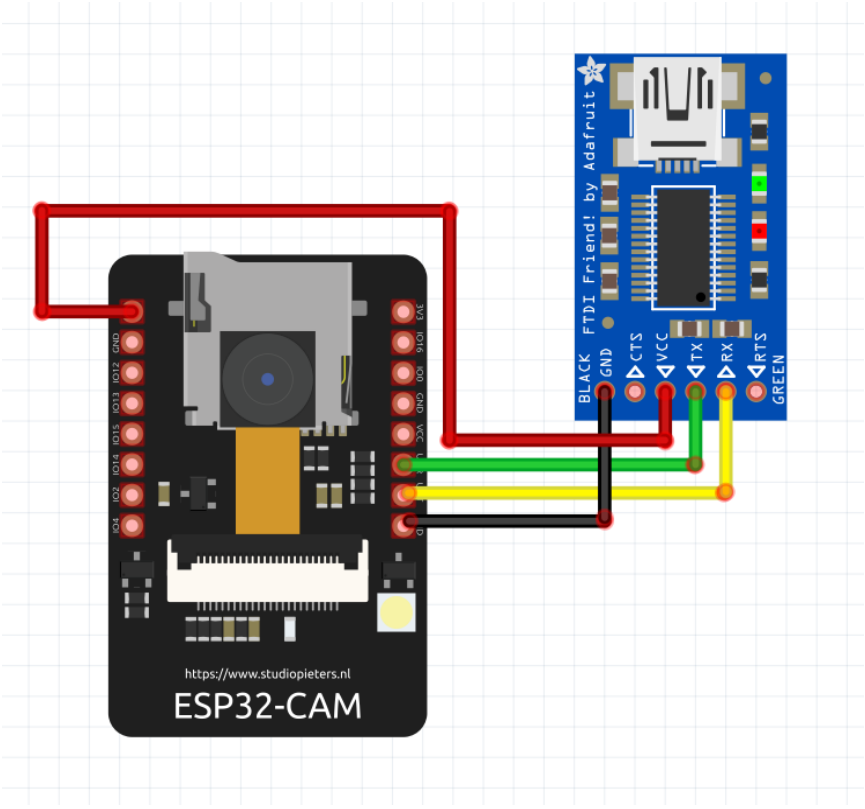
1. Firstly, connect the Camera Sensor with the FPC connector of ESP32-CAM , as shown:

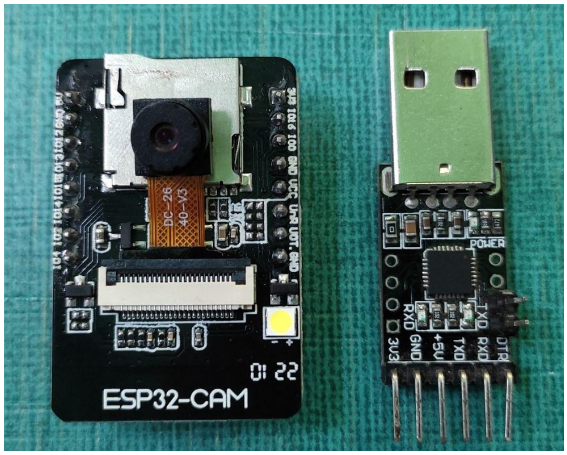


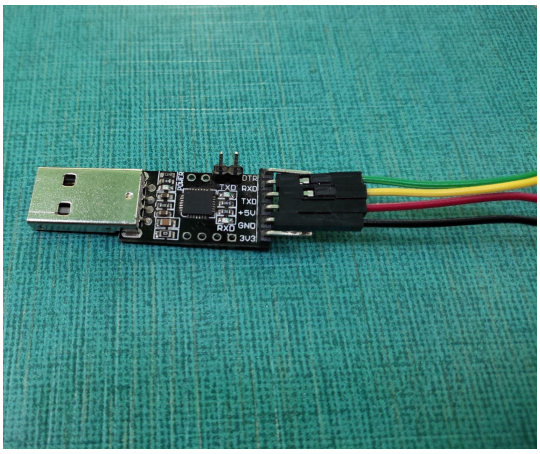


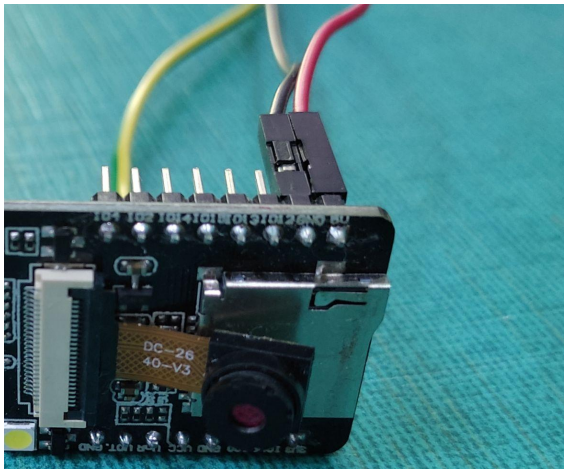


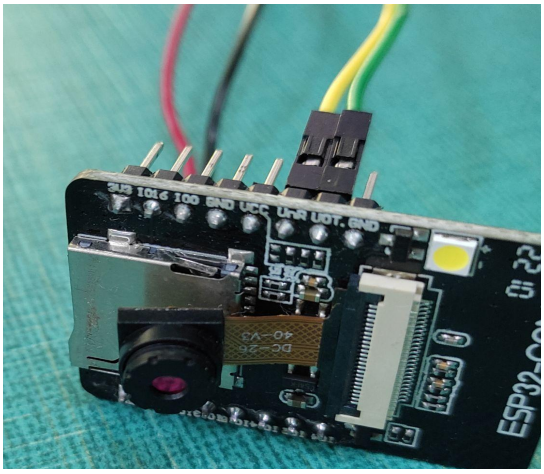
2. Connect ESP32 - CAM with USB-to-TTL converter by using female-to-female jumper wires, as shown:



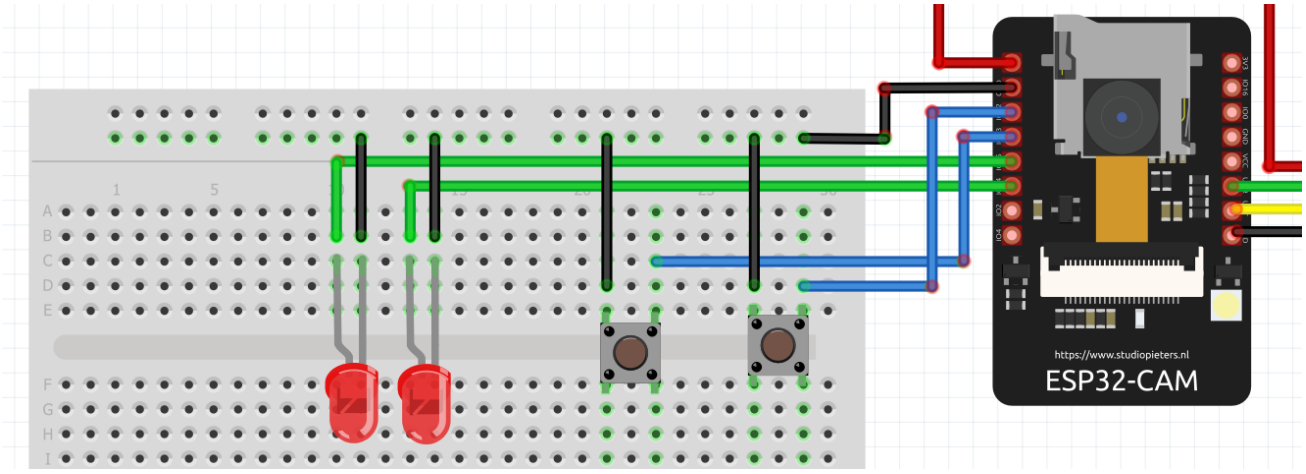








3. Connect Leds and Push Buttons with the above circuit, by using breadboard and jumper-wires.



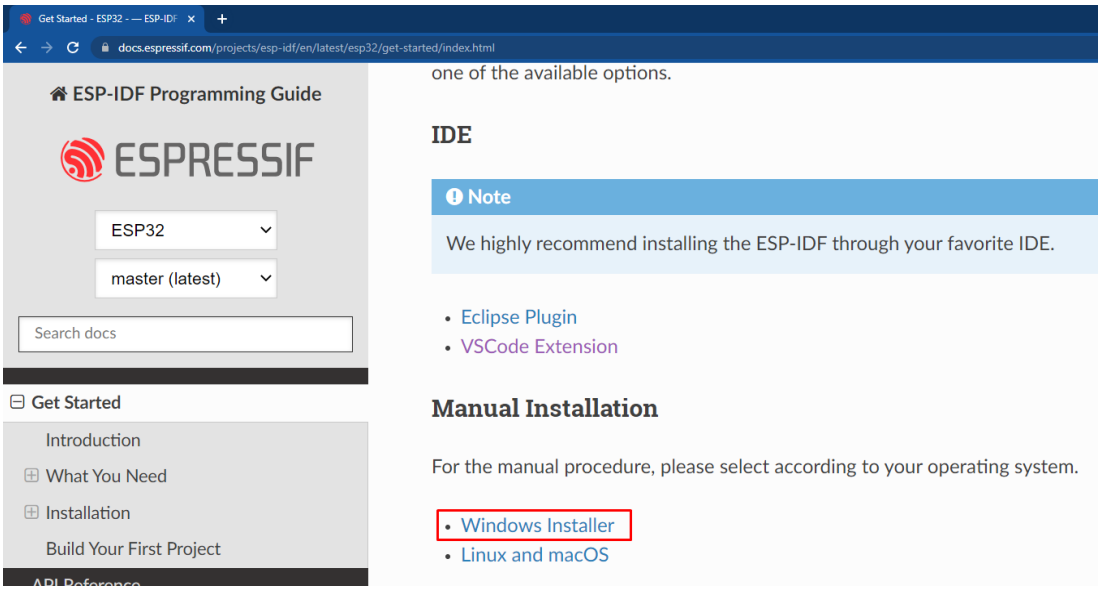
4. Finally, connect the USB-to-TTL converter with your PC.

SOFTWARE SETUP

- Install ESP-IDF in Windows
- Setting Up Firebase

1. ESP-IDF Installation in Windows:

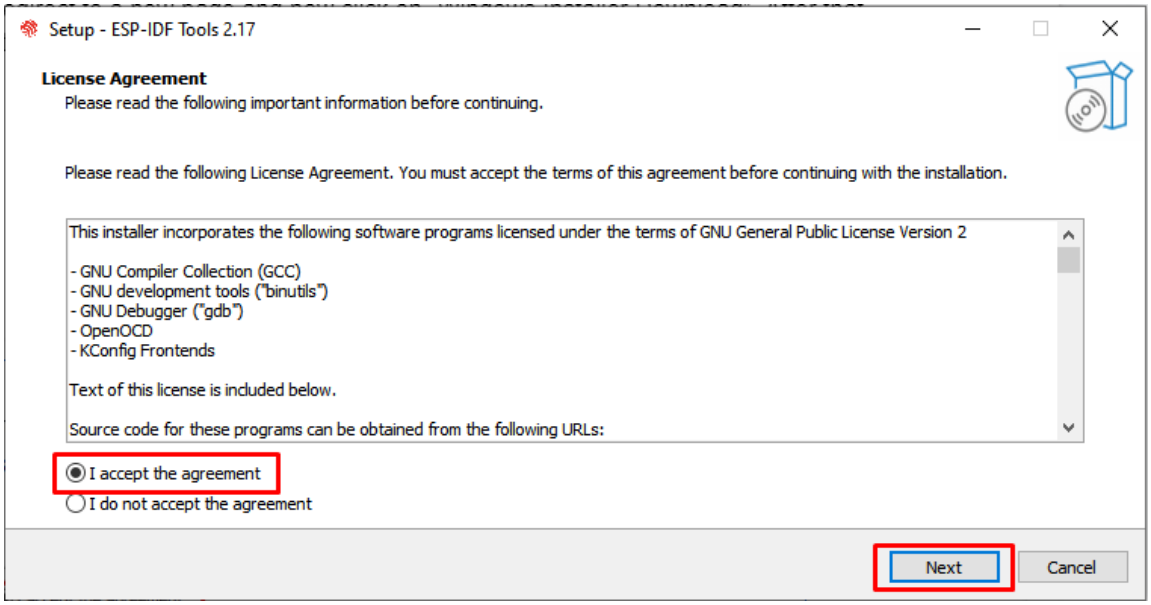
Opens the official ESP32 '[getting started](#)' webpage by Espressif. Scroll down until you reach the Manual Installation section. Here click '**Windows Installer**.'



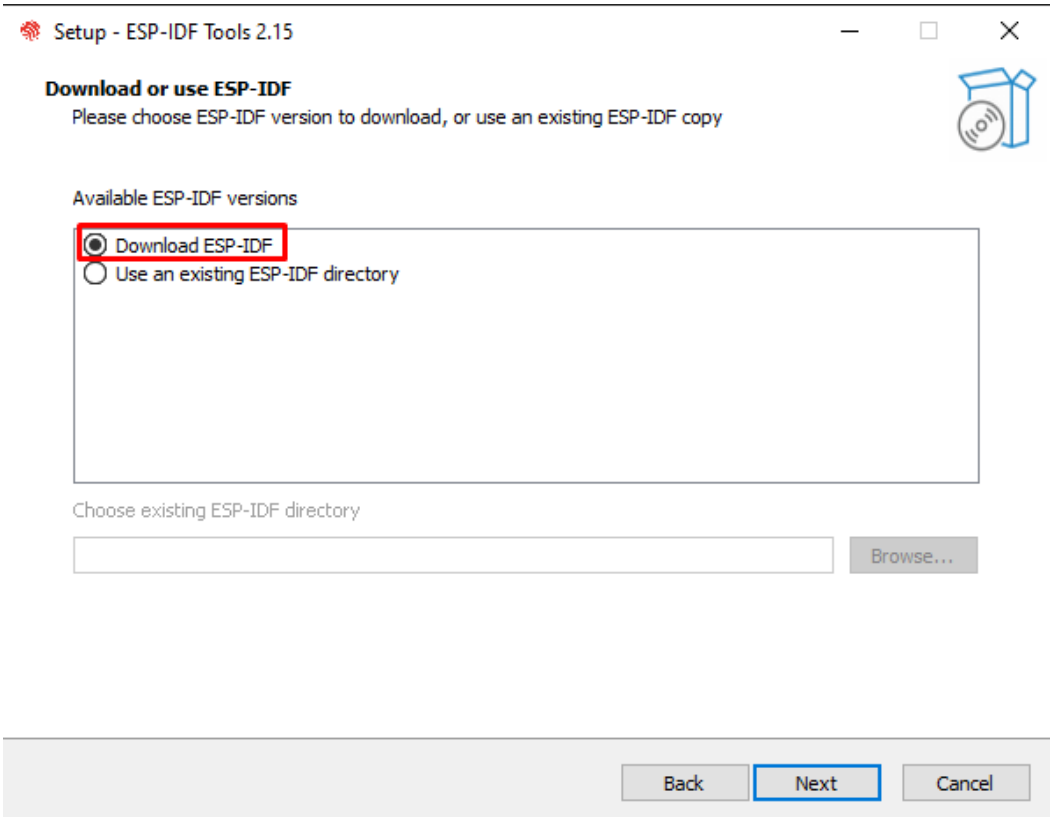
You'll redirect to a new page and now click on "**Windows Installer Download**". After that select the "**Universal Online Installer**".



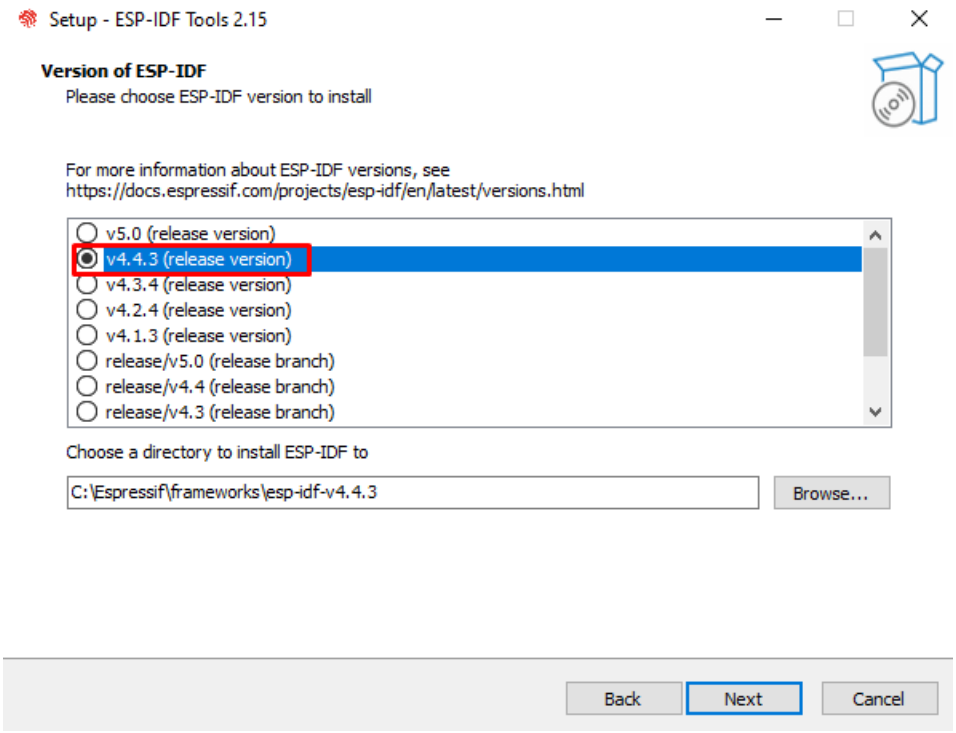
Now, open the downloaded .exe file, and begin esp-idf installation.



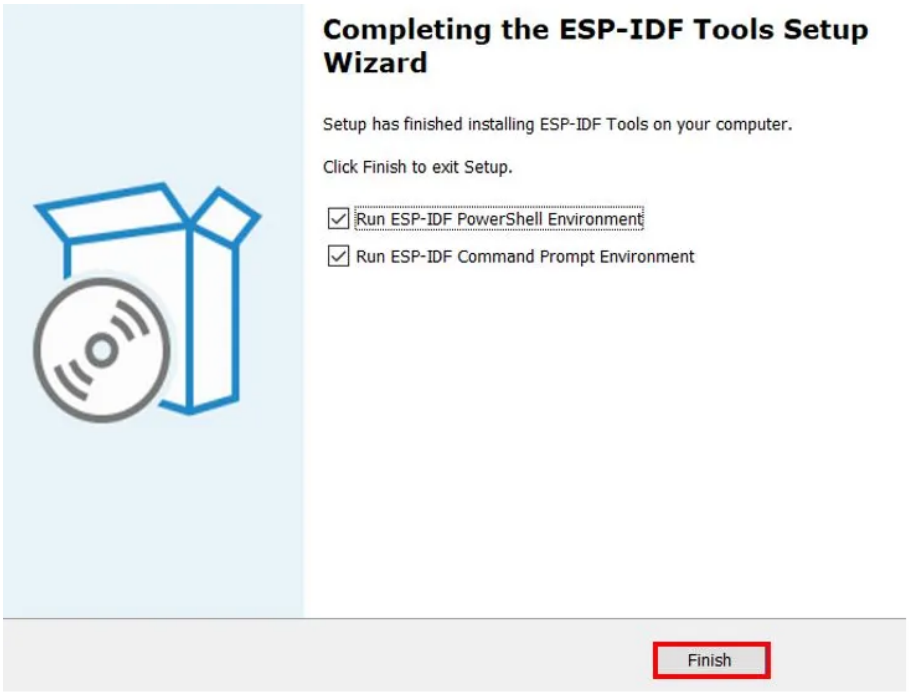
Select "**Download ESP-IDF**" and click "Next".



Select the “**V4.4.3**(release version)”.

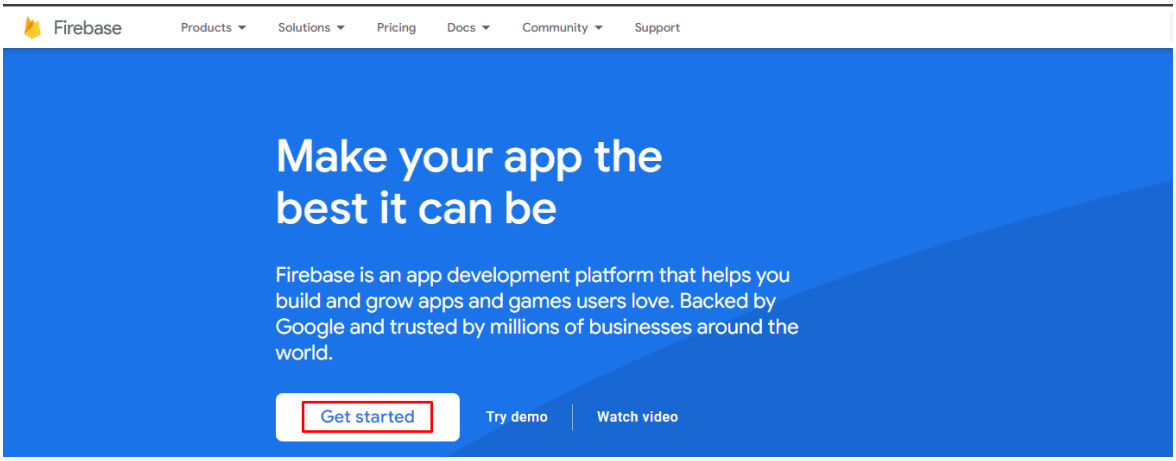


In the “Select Components” tab, remain it as it is and click “Next”. Now the installation begins. Click the **Finish button** to exit the setup.

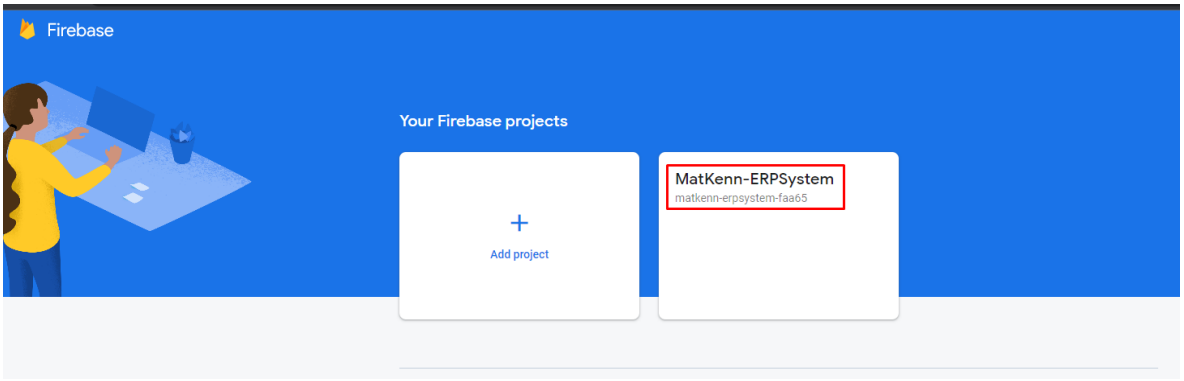


2. FireBase Setup:

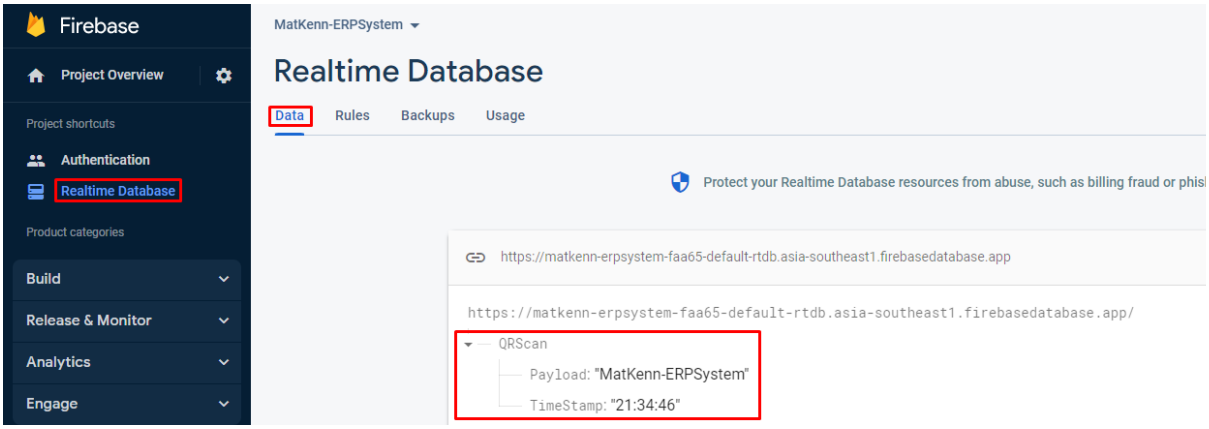
Now, setup the firebase by following these steps;
Open the official Firebase website and click on “**Get Started**”.



Sign-in with the provided credentials, and the “**Firebase Projects**” page will open.
Now click on “**MatKenn-ERPSystem**”.



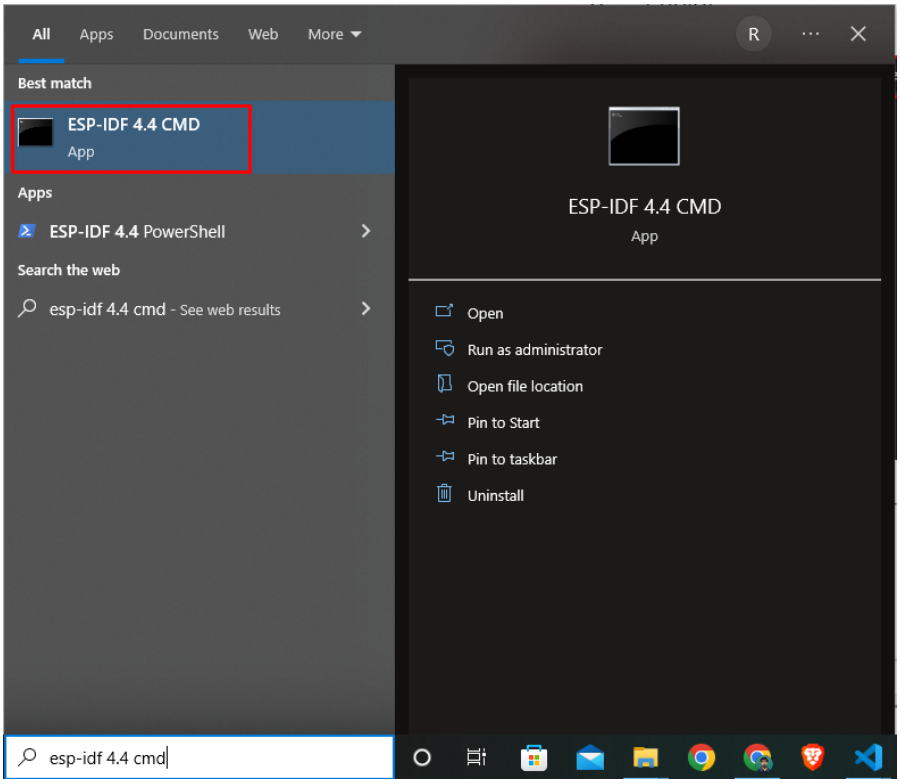
Go to **Realtime Database->Data**, and here you can access the qr-scanner data.



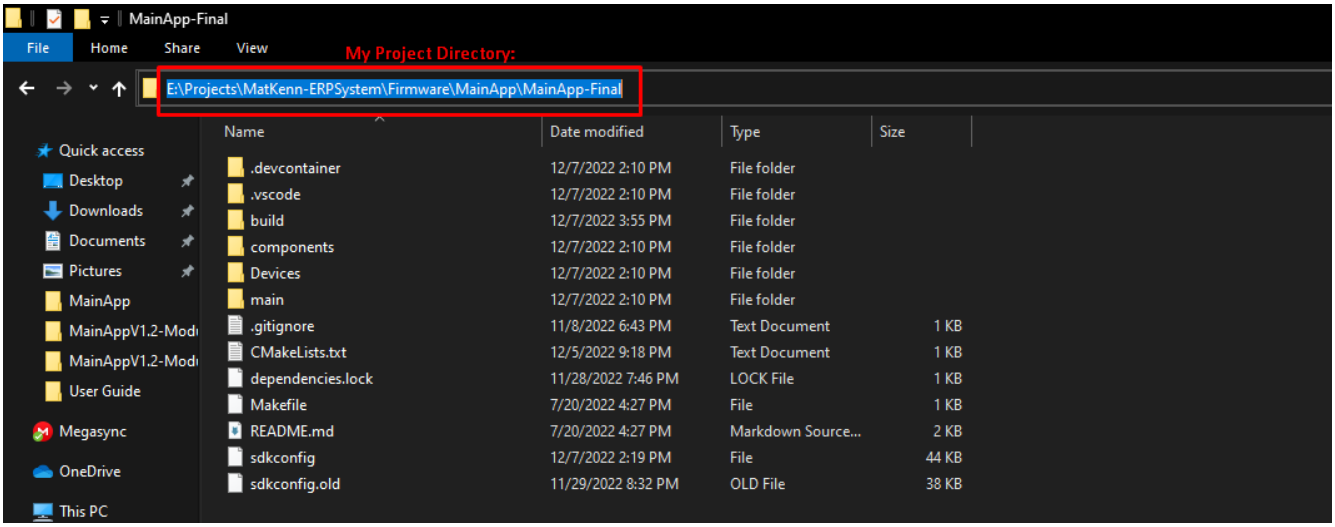
BUILDING, FLASHING & MONITORING (FIRMWARE):

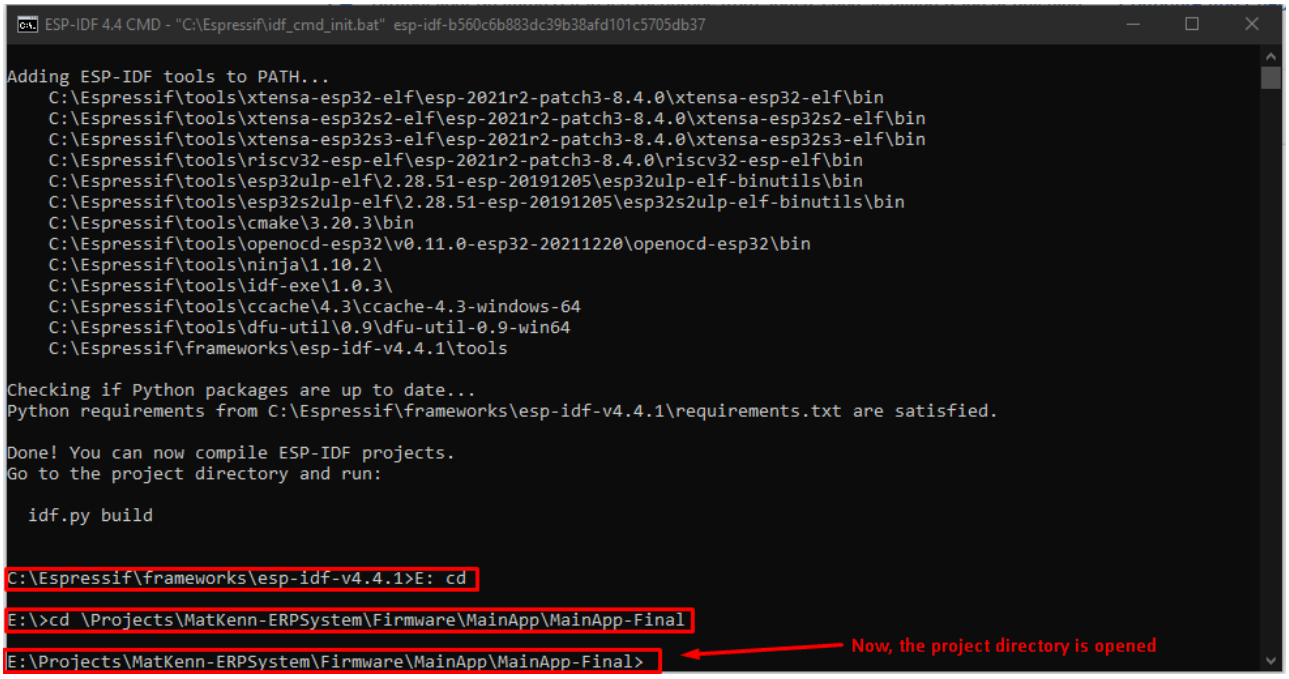
1. Project Build:

Open “**ESP-IDF CMD**” from the Windows start button.

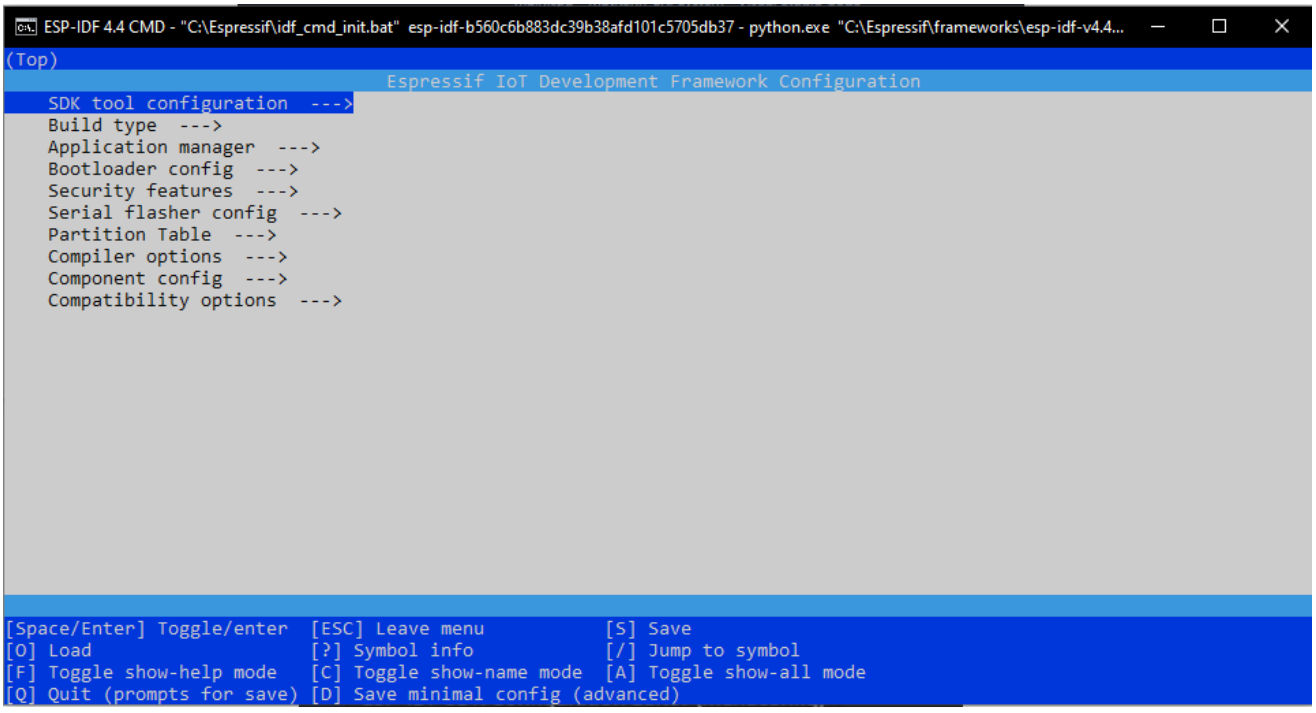


Now copy the project directory and open it in the ESP-IDF CMD;





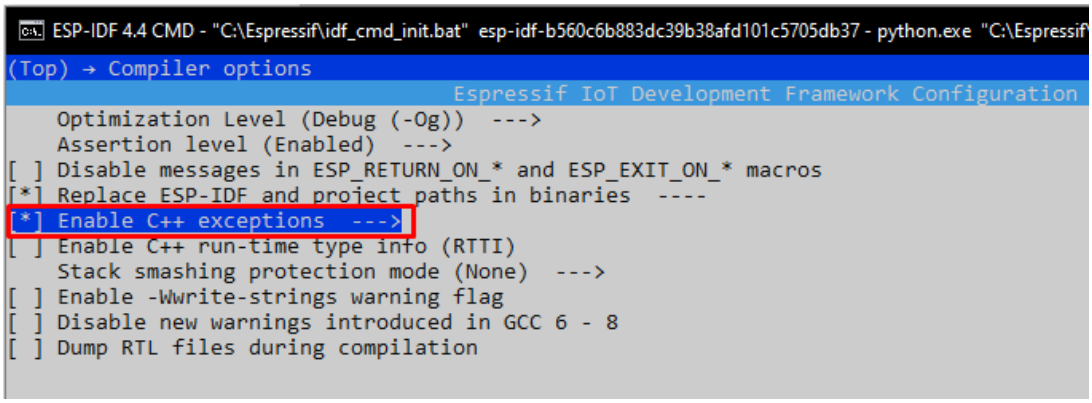
Open the “**menuconfig”** running the command; **idf.py menuconfig**
Change the following parameters & click save.



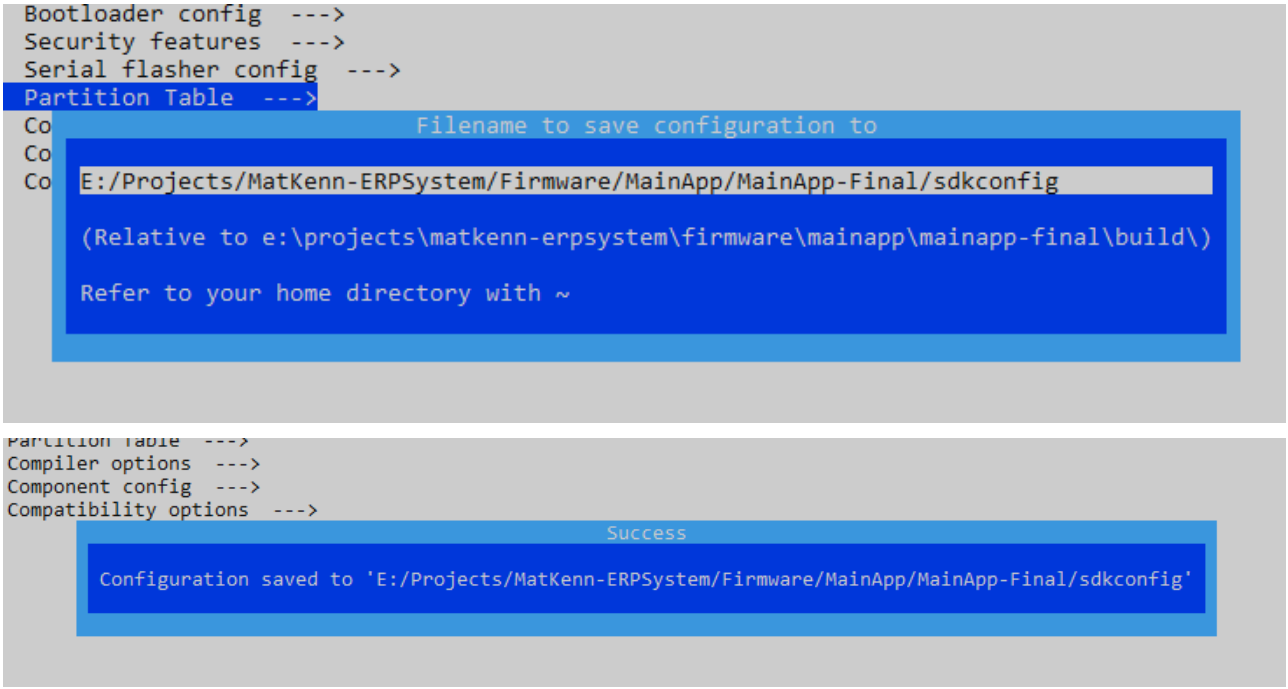
Go to the **Partition Table** and select "**Single factory app (large), no OTA**".



Go to the "**Compiler Options**" and select the "**Enable C++ exceptions**".
Set "Emergency Pool Size" to 0.

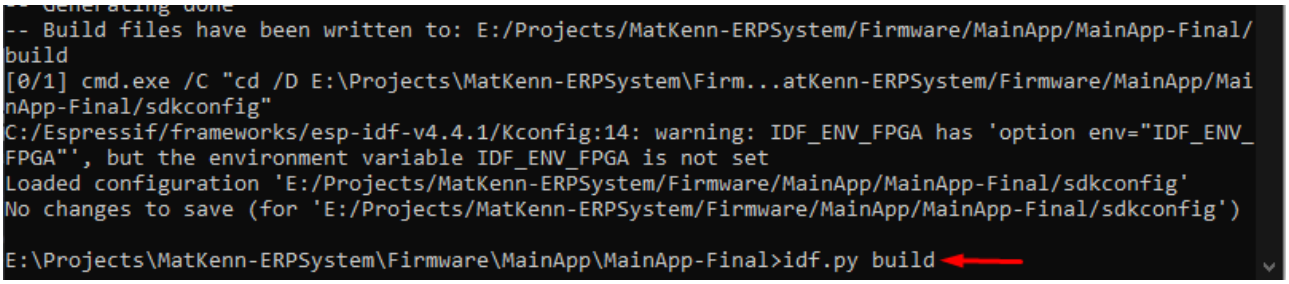


Press “S” and “Enter” to save the configuration.

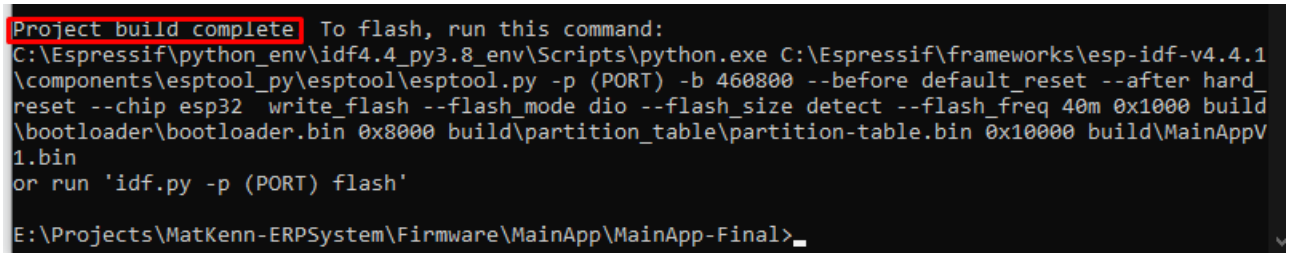


Press “ESC” to exit menuconfig.

Now, run the following command in the esp-idf terminal, and **Build the Project**.

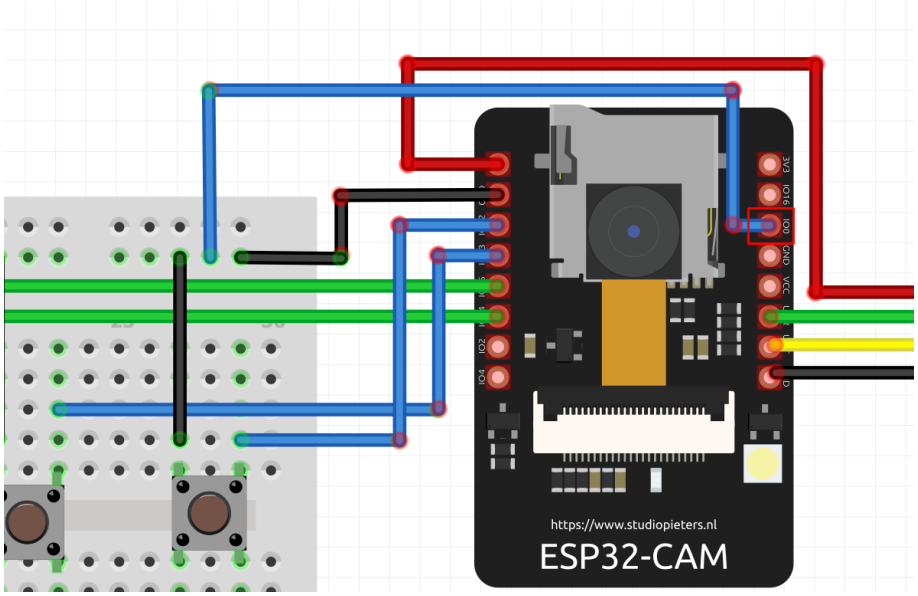


The project build is complete and ready to flash.

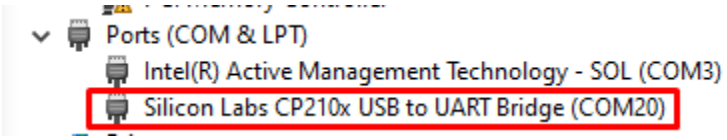


2. Flashing:

First you have to connect the **GPIO0** to GND and press the reset button of the ESP32-Cam board.



Goto **Device Manager->Ports** and check the COM Port.



Now, run this command on the esp-idf terminal, " **idf.py -p COM20 flash monitor**"

```
E:\Projects\MatKenn-ERPSystem\Firmware\MainApp\MainApp-Final>idf.py -p COM20 flash monitor
Executing action: flash
Running ninja in directory e:\projects\matkenn-erpsystem\firmware\mainapp\mainapp-final\build
Executing "ninja flash"...
```

Flashing is done. Remove "GPIO0" from "GND" and press the reset Button of the board.

```
Writing at 0x0013d504... (95 %)
Writing at 0x00142f8e... (97 %)
Writing at 0x001488ce... (100 %)
Wrote 1296944 bytes (732931 compressed) at 0x00010000 in 18.5 seconds (effective 561.6 kbit/s)...
Hash of data verified.
Compressed 3072 bytes to 105...
Writing at 0x00008000... (100 %)
Wrote 3072 bytes (105 compressed) at 0x00008000 in 0.1 seconds (effective 361.4 kbit/s)...
Hash of data verified.

Leaving...
Hard resetting via RTS pin...
Executing action: monitor
Running idf_monitor in directory e:\projects\matkenn-erpsystem\firmware\mainapp\mainapp-final
Executing "C:\Espressif\python_env\idf4.4_py3.8_env\Scripts\python.exe C:\Espressif\frameworks\esp
erpsystem\firmware\mainapp\mainapp-final\build\MainAppV1.2-Modular-espdl.elf -m 'C:\Espressif\pyth
--- WARNING: GDB cannot open serial ports accessed as COMx
--- Using \\.\COM20 instead...
--- idf_monitor on \\.\COM20 115200 ---
--- Quit: Ctrl+] | Menu: Ctrl+T | Help: Ctrl+T followed by Ctrl+H ---
```